

Hyperfinite dilation of Markov semigroups: base-algebra reduction and a compatible-root criterion

A partial answer to Remark 3.10 of arXiv:1603.04901

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Abstract

Arhancet asks whether a weak* continuous semigroup of hyper-factorizable Markov maps always has a dilation on a hyperfinite von Neumann algebra. We prove three structural results. First, the hypothesis itself forces the base algebra to be semidiscrete; with separable predual it is therefore hyperfinite. This remains true if hyper-factorizability is assumed only for positive times. Second, every finite set of nonnegative rational times has an exact simultaneous hyperfinite dilation, so no finite rational fragment can witness failure. Third, we give a sufficient condition that completes the program: compatible hyperfinite discrete dilations along the factorial meshes $1/n!$ have a hyperfinite inductive limit, whose rational action extends to the required continuous real action. The unsolved step is exactly the construction of one infinite compatible tower. Arbitrarily long finite towers exist, but the ultraproduct compactness used in the source does not preserve hyperfiniteness.

1 The question and conventions

Let M be a von Neumann algebra with normal faithful state ϕ . A ϕ -Markov map $T : M \rightarrow M$ is *hyper-factorizable* if there are a hyperfinite von Neumann algebra P with normal faithful state χ and (ϕ, χ) -Markov embeddings $J_0, J_1 : M \rightarrow P$ such that

$$T = J_0^* J_1.$$

Here J_0^* is the adjoint Markov map, equivalently the state-preserving conditional expectation onto $J_0(M)$ followed by J_0^{-1} .

A weak* continuous semigroup $(T_t)_{t \geq 0}$ is *hyper-dilatable* if there are a hyperfinite (N, ψ) , a (ϕ, ψ) -Markov embedding $J : M \rightarrow N$, and a weak* continuous state-preserving group $(U_t)_{t \in \mathbb{R}} \subset \text{Aut}(N)$ such that

$$T_t = J^* U_t J, \quad t \geq 0. \tag{1}$$

Remark 3.10 of Arhancet [1] asks whether pointwise hyper-factorizability implies (1).

We use two standard preservation facts. Hyperfinite von Neumann algebras are semidiscrete. Also, the discrete dilation construction for a hyper-factorizable Markov map can be carried out through the hyperfinite factorization algebra and countable state-preserving inductive limits; hence it yields a hyperfinite discrete dilation. This is the hyperfinite version of the factorization–dilation construction in [2, Theorem 4.4] and is also the finite-mesh input behind Remark 3.9 of [1].

2 The base algebra is already semidiscrete

Proposition 1 (Hyperfinite-base reduction). *Let $(T_t)_{t \geq 0}$ be weak* continuous at 0. If T_t is hyperfactorizable for every $t > 0$, then M is semidiscrete. In particular, if M has separable predual, then M is hyperfinite.*

Proof. We prove the finite-set form of semidiscreteness. Fix a finite set $F \subset M$, a finite set $\Omega \subset M_*$, and $\varepsilon > 0$. Weak* continuity gives $t > 0$ such that

$$|\omega(T_t(x) - x)| < \varepsilon/2 \quad (x \in F, \omega \in \Omega). \quad (2)$$

Choose a hyperfinite factorization

$$M \xrightarrow{J_1} P \xrightarrow{J_0^*} M, \quad T_t = J_0^* J_1. \quad (3)$$

Since P is semidiscrete, there are a matrix algebra A and normal ucp maps $a : P \rightarrow A$ and $b : A \rightarrow P$ such that

$$|\omega(J_0^* b a J_1(x) - J_0^* J_1(x))| < \varepsilon/2 \quad (x \in F, \omega \in \Omega). \quad (4)$$

The ucp map $J_0^* b a J_1 : M \rightarrow M$ factors through the matrix algebra A . Combining (2)–(4) shows that it approximates id_M on F in every functional in Ω within ε . Directing these choices by (F, Ω, ε) gives a point-ultraweak approximation of id_M by normal ucp maps through matrix algebras. Thus M is semidiscrete. For separable predual, semidiscreteness is equivalent to injectivity and to hyperfiniteness. $\square$

Remark 2. *If the phrase “every T_t ” includes $T_0 = \text{id}_M$, there is a shorter argument: a hyperfinite factorization of id_M makes M a ucp retract of a hyperfinite algebra. Proposition 1 is stronger because it uses only positive times and continuity at zero.*

3 Every finite rational fragment is hyper-dilatable

Proposition 3 (Exact local compatibility). *Assume every T_t , $t > 0$, is hyperfactorizable. Given a finite set $F \subset \mathbb{Q}_+$, there are a hyperfinite (N_F, ψ_F) , a Markov embedding $J_F : M \rightarrow N_F$, and a state-preserving automorphism S_F such that*

$$T_q = J_F^* S_F^{dq} J_F \quad (q \in F) \quad (5)$$

for some positive integer d with $dq \in \mathbb{N}$ for all $q \in F$.

Proof. Choose a common denominator d . The map $T_{1/d}$ is hyperfactorizable, so its discrete factorization–dilation gives a hyperfinite (N_F, ψ_F, J_F, S_F) satisfying

$$J_F^* S_F^k J_F = (T_{1/d})^k = T_{k/d} \quad (k \geq 0).$$

Taking $k = dq$ proves (5). $\square$

Thus all relations involving finitely many rational times can be realized in one hyperfinite algebra. The source’s ultraproduct turns these locally compatible models into one global rational action. The obstruction is that hyperfiniteness, unlike QWEP, is not preserved by that ultraproduct step.

4 A compatible-root criterion for the full conclusion

Put $h_n = 1/n!$. A *factorial dilation tower* consists of hyperfinite von Neumann algebras (N_n, ψ_n) with separable preduals, Markov embeddings $J_n : M \rightarrow N_n$, state-preserving automorphisms S_n , and Markov embeddings $\iota_n : N_n \rightarrow N_{n+1}$ such that

$$J_n^* S_n^k J_n = T_{kh_n} \quad (k \geq 0), \quad (6)$$

$$\iota_n J_n = J_{n+1}, \quad (7)$$

$$\iota_n S_n = S_{n+1}^{n+1} \iota_n. \quad (8)$$

Theorem 4 (Compatible-root completion). *Suppose M has separable predual and $(T_t)_{t \geq 0}$ admits a factorial dilation tower. Then the semigroup is hyper-dilatable.*

Proof. Take the state-preserving von Neumann inductive limit (N_∞, ψ) of the system (N_n, ι_n) . We identify each N_n with its image. The modular compatibility of Markov embeddings gives state-preserving conditional expectations between the stages. Since the tower is countable, has separable preduals, and each stage is hyperfinite, N_∞ is hyperfinite.

Relations (8) define a state-preserving action $U : \mathbb{Q} \rightarrow \text{Aut}(N_\infty)$. Indeed, for $q \in \mathbb{Q}$ and $x \in N_m$, choose $\ell \geq m$ with $q/h_\ell \in \mathbb{Z}$ and set

$$U_q(x) = S_\ell^{q/h_\ell}(x), \quad (9)$$

where x is viewed in N_ℓ . Since $h_{\ell+1} = h_\ell/(\ell+1)$, relation (8) makes (9) independent of ℓ . It also gives $U_{q+r} = U_q U_r$.

The embeddings J_n induce a Markov embedding $J : M \rightarrow N_\infty$. For $q \in \mathbb{Q}_+$, choose n with $q/h_n \in \mathbb{N}$. Equations (6) and (9) give

$$J^* U_q J = T_q. \quad (10)$$

Moreover, for $x \in M$ and $q \downarrow 0$ through $\mathbb{Q}_+$,

$$\|U_q J(x) - J(x)\|_{2,\psi}^2 = 2\phi(x^*x) - 2\text{Re } \phi(x^* T_q x) \longrightarrow 0. \quad (11)$$

Applying (11) to x^* gives strong* continuity at zero on $J(M)$; negative q follow by invariance. Hence the same is true on every rational translate $U_r J(M)$.

Let

$$N_0 = (U_q J(M) : q \in \mathbb{Q})'' \subset N_\infty.$$

On bounded sets, the strong* topology is compatible with multiplication, and the continuous vectors of an isometric group representation form a closed subspace. It follows first on the algebra generated by the translates and then by strong* closure that $q \mapsto U_q(x)$ is strong* continuous for every $x \in N_0$. Thus $U|_{N_0}$ extends uniquely to a weak* continuous group $(\tilde{U}_t)_{t \in \mathbb{R}}$ of state-preserving automorphisms.

The algebra N_0 has separable predual and is invariant under the modular group of ψ , because J is Markov and every U_q preserves ψ . Takesaki's theorem therefore supplies a normal conditional expectation $N_\infty \rightarrow N_0$. Hence N_0 is injective; separability makes it hyperfinite. Finally, weak* continuity of both sides of (10), together with density of $\mathbb{Q}_+$ in $\mathbb{R}_+$, yields

$$T_t = J^* \tilde{U}_t J \quad (t \geq 0).$$

This is the required hyperfinite dilation. □

5 How close the criterion is to being automatic

For every m , choose a hyperfinite discrete dilation of $T_{1/m!}$. Restricting its shift to the powers

$$1, \quad m, \quad m(m-1), \quad \dots, \quad m!$$

produces a compatible tower of every prescribed finite length. Therefore all finite subsystems of (6)–(8) are satisfiable. What is missing is one infinite branch. Ordinary compactness supplies such a branch only in an ultraproduct, and a tracial ultraproduct of hyperfinite factors can be noninjective. Likewise, weak* limits of finite-dimensional moment models can have noninjective GNS closures. These facts explain both why the source’s QWEP argument works and why it does not settle the hyperfinite question.

Theorem 4 reduces the open problem to a concrete lifting issue: choose the discrete dilations of the successive roots so that the coarse shift is exactly the appropriate power of the fine shift under a Markov embedding. No finite-time obstruction remains.

6 Scope and novelty guardrail

The packet does not claim a full answer to Remark 3.10. Its unconditional content is Proposition 1 and Proposition 3; Theorem 4 is a rigorous completion criterion whose compatibility hypothesis is the remaining gap.

Eight focused upgrade attempts were made. They tested direct preservation of injectivity in the source ultraproduct, nested dyadic/factorial dilations, compact path-state models, product-system ideas, and counterexamples based on noninjective minimal path algebras. The detailed attempt log records why each unresolved route returns to the same coherence obstruction.

On 11 August 2026, bounded novelty searches covered the run’s cheap indexes, the exact question and terminology, and all 17 citing works reported by OpenAlex for the published paper. They found special positive results for Schur and Fourier multipliers and work on individual absolute dilations, but no later full answer to Remark 3.10 and no occurrence of the semidiscrete-base reduction above. This is a novelty guardrail, not a priority claim.

Human-review recommendation. Review as a substantial partial result. The main checks are: (i) the matrix-factorization net in Proposition 1; (ii) preservation of hyperfiniteness in the standard discrete dilation of one hyper-factorizable map; and (iii) the strong*-continuous extension and expected-subalgebra step in Theorem 4.

References

- [1] C. Arhancet, *Dilations of semigroups on von Neumann algebras and noncommutative L^p -spaces*, J. Funct. Anal. **276** (2019), 2279–2314; arXiv:1603.04901.
- [2] U. Haagerup and M. Musat, *Factorization and dilation problems for completely positive maps on von Neumann algebras*, Comm. Math. Phys. **303** (2011), 555–594; arXiv:1009.0778.
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